



CANDIDATE
NAME

--

CIVICS
GROUP

1	7	-		
---	---	---	--	--

REGISTRATION
NUMBER

--	--

H2 Biology

Paper 4 Practical

9744/04

21 August 2018

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use

Appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
Section A	
1	
2	
3	
Total	55

Answer **all** the questions.

- 1** Yeast cells contain enzymes which catalyse the breakdown of glucose to produce carbon dioxide and water.

The carbon dioxide reacts with water and forms a weak acid.

Bromothymol blue is a pH indicator and changes colour as shown in Table 1.1.

Table 1.1

pH	colour of bromothymol blue
8	Blue
7	Green
6	Yellow

You are required to:

- immobilise yeast cells in sodium alginate beads
- find the mean surface area of the sodium alginate beads
- investigate the independent variable, changing the surface area of the beads, on rate of enzyme reaction

You are provided with:

- yeast suspension, labelled **Y**
- sodium alginate solution, labelled **S**
- calcium chloride solution, labelled **C**
- glucose solution, labelled **G**
- bromothymol blue, labelled **B**
- sodium hydroxide solution, labelled **A**

Proceed as follows.

- 1** Put 20 cm³ of **C** into a boiling tube.
- 2** Put 5 cm³ of **S** into a small beaker or container.
- 3** Collect 5 cm³ of **Y** from below the froth and put it into the same container as **S**. Mix well.
- 4** Use a 5 cm³ syringe to collect 2 cm³ of the mixture **S** and **Y**.

- 5 Suspend the 5 cm³ syringe over the boiling tube containing **C** as shown in Fig. 1.1.

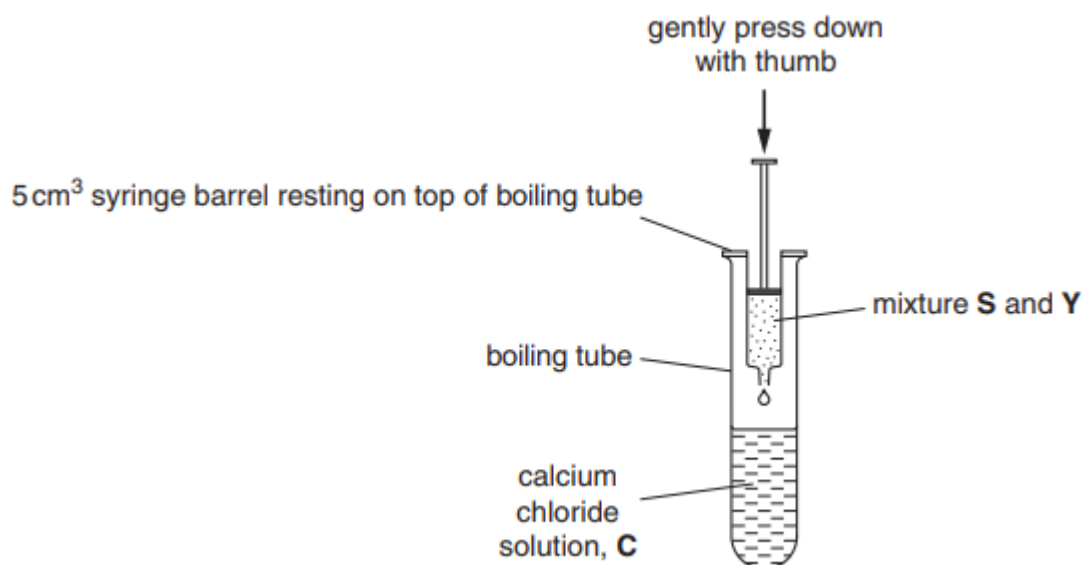


Fig. 1.1

- 6 Gently press down on the plunger of the syringe with your thumb to release a drop into solution **C**. The drop should form a bead.
- 7 Repeat step 6 to make the number of beads that you think you will need.
- 8 Tip the contents of the boiling tube into a Petri dish. You may use blunt forceps to pick up the beads as you select them for use in the experiment.

You will need to calculate the mean surface area of the beads.

To do this,

- decide on the number of beads you will measure
- use the 2 mm × 2 mm grid to measure each bead
- calculate the surface area of each bead using the formula
surface area = $4\pi r^2$ where $\pi = 3.14$, r = radius of a bead
- calculate the mean surface area of the beads.

In the space below, draw a bead and indicate clearly on your drawing the part of the bead you will be measuring with the 2 mm x 2 mm grid.

- (a) (i) Prepare the space below to show your measurements and calculations.

Show all the steps in your calculation of the mean.

1. 5 or more beads radii / diameters used + units in mm;
(R: any measurements for diameter not whole number)
(R: more than 6mm for diameter, or more than 3mm for radius)
2. Correct calculation for surface area of one bead (inclusive of working);
3. Working shows each surface area added together and divided by number of beads; (note: include worked examples to show diff between standard ans and ans using average bead radius to calculate avg bead S.A)
4. Final answer no more than 3 s.f.;

mean surface area of the beadsmm² [4]

Before starting the investigation, read through steps **9 – 15** (including **(a)(ii)**) and prepare a table in **(a)(iii)**.

A student suggested that it was possible to investigate the independent variable, surface area, by changing the number of beads. The maximum number of beads used was 20. Decide the other numbers of beads to use and state the different number of beads you will use.

(ii)
.....[1]

1. Equal interval and at least 3 different numbers of beads, inclusive of 20;
(A: if 20 is missing from this part but appears in next part)

- 9 Label as many test-tubes as you will need with the number of beads for each test-tube.
- 10 Put 10 cm³ of solution **G** into each test-tube.
- 11 Put 1 cm³ of **B** into each test-tube. Attach the rubber bung to each test-tube. Mix the contents in the test tube by inverting the test tube a few times.
- 12 If the contents of the test-tube are not blue, add one drop at a time of **A** to the contents of each test-tube to turn them all the same blue colour. Mix well after adding each drop.
- 13 Put the required number of beads into each test-tube.
- 14 Attach the rubber bung to each test-tube. Turn and mix the contents. Mix every 2 minutes for 6 minutes.
- 15 Record your observations at 2 minutes interval, up to 6 minutes.

(ii) Prepare the space below to record your observations.

CONSTRUCT TABLE

1. Table with all cells drawn + headings: (surface area / mm²) / (no. of beads) and (colour or observation);
(R: units in body of table)
(R: if heading is incorrect but title of table is correct)
(R: soln)
2. Only records at 2, 4 and 6 minutes; (Ignore if observation at 0 min recorded.
R: if observations for more than 6 min recorded)
3. Use 20 beads in one tube and at least 3 different number of beads (of equal intervals); (ecf from prev part if recorded same no. of beads as mentioned in prev part)
4. Trend: Highest number of beads / surface area (turns yellow the fastest) + only allow yellow to be recorded for highest no. of beads as at least 12 beads; (A: green if highest no. of beads used is a small number) (A: if highest two to three number of beads same result, if no. of beads used is low)

[4]

- (iii) The student realised that there were two independent variables in this procedure.

Other than surface area or number of beads, state the **other** independent variable.

.....
[1]

1. Yeast / enzyme concentration or quantity/volume;

- (iv) Suggest a suitable control that could have been used in this investigation.

.....
[1]

1. Keeping all other variables constant, use boiled and cooled yeast / water instead to show that the colour change of the indicator is due to respiration by yeast;

R: use alginate without yeast

R: boiled yeast (without mention of cooling)

R: setup without beads

R: ... to show that glucose breakdown is due to yeast

- (v) Identify an error and suggest a corresponding improvement to the procedure to be carried out to improve the confidence in your results.

.....

[2]

1. Yeast alginate beads created using the syringe are not of equal size / diameter, hence surface area / volume tested may not be accurate;

2. Cut cubes of equal size/dimensions from a block of yeast-alginate. Each cube will contain equal volume of yeast;

A: micropipette to dispense fixed vol of yeast to create beads containing constant volume of yeast

OR

3. Using human eye to determine colour change is subjective;

4. Use a colorimeter to detect colour change;

OR

5. Temperature was not kept constant during the experiment;

6. Use a thermostatically-controlled water bath to maintain temperature at e.g. 30°C;

A: 25 to 45°C

OR

7. No replicates were conducted;

8. Conduct two more replicates to check for anomalous results / find mean or average;

OR

9. Inaccurate observations made at the various time intervals as experiment started at the same time for all surface areas / number of beads tested;
10. Stagger start or have separate experiments for each surface area tested;

OR

(If used less than 5 different number of beads)

11. Range of values tested for independent variable too few;
12. Use more beads or more surface areas;

- (b) In another experiment, a students investigated the effect of varying the size of alginate beads on the rate of breakdown of hydrogen peroxide by plant catalase.

Plant extract containing catalase can be immobilised using alginate beads as well.

Other variables were considered and kept constant. There were 10 beads of each diameter.

Fig. 1.2 shows two of the sets of beads and the distribution of the plant catalase.

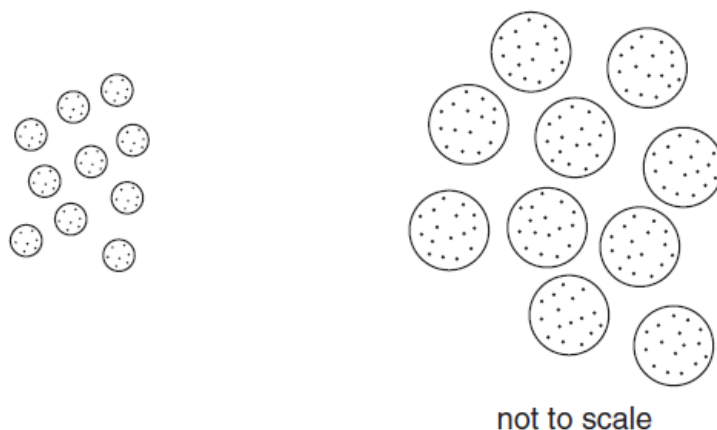


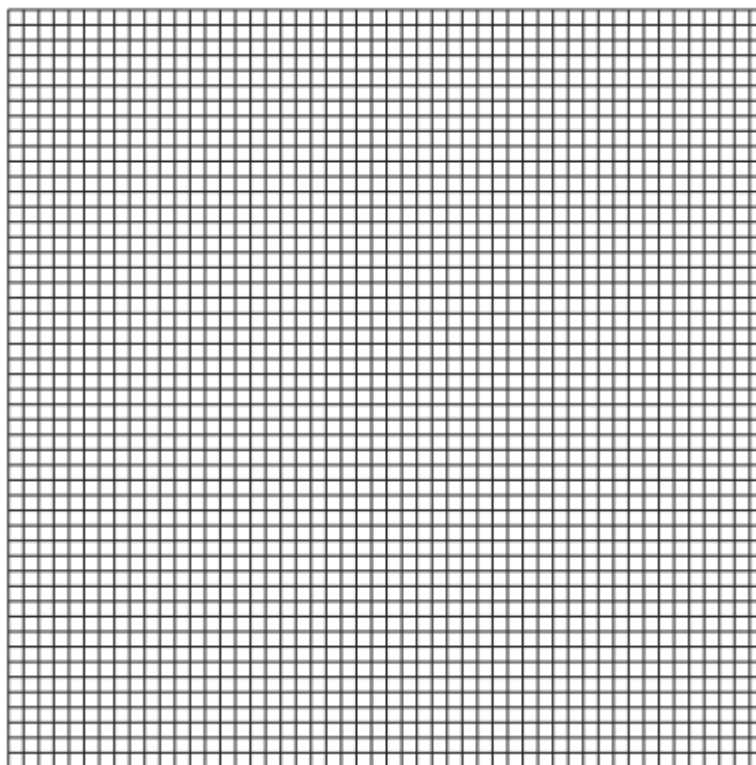
Fig. 1.2

The results of the student's investigation are shown in Table 1.2.

Table 1.2

bead diameter /mm	mass of product in a minute /mg min ⁻¹
1.0	24.0
1.5	30.5
3.0	42.0
4.5	50.0
6.0	52.5

- (i) Plot a graph of the data shown in Table 1.2.



1. (S) x-axis scale (1.0mm to 2cm labelled each 2cm, except origin and 6.0) + y-axis scale (10 mgcm⁻³ to 2 cm labelled each 2cm except origin and 60) + intervals marked out for both axes
R: wrong orientation
2. (A) x-axis (bead diameter / mm) + y-axis (mass of product in a minute / mg min⁻¹)
3. (P) correct plotting of 5 points as small cross or dot in circle or cross in circle to within half a square;
R: plotted 2cm with same symbol as other points
R: dots alone
R: if dot (in circle) bigger than 1mm diameter
4. (LE) five plots with ruled lines exactly point to point OR curve through 5 points AND (quality) smooth line less than 1mm thick;
ECF for wrong plots
R: extrapolation

[4]

- [4]

- [Total: 21]

- 2 (a) Two single-celled algal species, *Scenedesmus subspicatus* and *Chlorella vulgaris* are common in polluted waters and can be used in sewage lagoons to reduce the nitrate content of the water.

A student wanted to determine the minimum amount of light that needs to be provided to each of the two types of algae for maximum growth to occur.

Design an experiment to estimate the light saturation points of both algae.

In your plan, you must use:

- a growing culture of *Chlorella vulgaris*
- a growing culture of *Scenedesmus subspicatus*
- a range of different neutral density filters
(Note: these filters each allow different percentages of light to pass through, e.g. a 10% neutral density filter allows 10% of light to be transmitted)
- a lamp that does not emit heat
- bromothymol blue indicator

You may select from the following sterilised apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- syringes
- timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include the layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific forms
- include reference to safety measures to minimise the any risks associated with the proposed experiment.

[9]

A. Aim (no mark)

To determine if *Scenedesmus subspicatus* or *Chlorella vulgaris* has a lower light saturation point

B. (max 2)

- Independent variable = light intensity, as varied by using different neutral density filters (e.g. 20%, 40%, 60%, 80% and 100% light transmission);
- Measurable variable = time taken for bromothymol blue to turn from green to blue
- Dependent variable = rate of photosynthesis as determined by $1/(\text{time taken for bromothymol blue to turn from green to blue})$; (measurable variable + dependent variable = 1 mark)

C. Theory (max 2)

1. Trend = as light intensity increases till light saturation point, the rate of bromothymol blue turning from green to blue increases;
2. This is because as light intensity increases, rate of photosynthesis increases and the rate of assimilation of carbon dioxide by algae increases, increasing pH of indicator faster
3. Trend = Beyond light saturation point, increase in light intensity does not increase rate of bromothymol blue turning from green to blue;
4. This is because rate of photosynthesis is at maximum and other factors such as concentration of algae / temperature is limiting

D. Pilot test (no mark)

Conduct a pilot experiment to determine suitability of apparatus, optimum conditions, amount of materials and range of independent variable used.

E. Procedure (1 mark for using all required apparatus and reagents correctly; 2 algae + ND filters + lamp + indicator)

1. Using a syringe, put 10 cm³ of *Chlorella vulgaris* culture into a test tube.
2. Wrap the test tube containing 10 cm³ of *Chlorella vulgaris* culture with a 20% neutral density filter.
3. Using a syringe, put 5 cm³ of bromothymol blue indicator into another test tube.
4. Use two rubber bungs and a delivery tube to connect the test tubes containing *Chlorella vulgaris* and bromothymol blue. Make sure the connection is secure.
5. Place the test tube containing *Chlorella vulgaris* 5 cm away from the lamp.
6. Start the stopwatch.
7. Stop the stopwatch once bromothymol blue has changed from green to blue and record the time taken.
8. Conduct two more replicates using the 20% neutral density filter, to check for anomalous results. (1 mark)
9. Repeat steps 1 to 8 using the other neutral density filters that allow 40%, 60%, 80% and 100% light transmission.
10. Include a control in experiment. Keeping all other variables constant, replace the *Chlorella vulgaris* used with boiled and cooled *Chlorella vulgaris* / water. This is to show that presence of the algae is responsible for colour change of bromothymol blue. (1 mark)
11. Repeat the entire experiment two more times to check for reproducibility of results. (1 mark)

12. Constants (max 2):

- Using a syringe, volume of bromothymol blue to be kept constant 5 cm³ for all setups. The volume of indicator used will influence the time taken for indicator to change from green to blue / more indicator used will increase the time taken for colour change from green to blue (ORA);
- distance from lamp to be kept constant at 5 cm using a ruler for all setups. Distance from lamp will change light intensity and will affect the rate of photosynthesis.
- Concentration of *Chlorella vulgaris* to be kept constant. Determine the concentration of algae via cell counting and dilute the algae corresponding to a constant concentration. *Chlorella vulgaris* carries out photosynthesis and therefore the number of cells will affect rate of photosynthesis and hence rate of colour change of indicator;
- Temperature to be kept constant at 30°C, using a thermostatically controlled water bath / monitor temperature using a thermometer and maintaining the temperature at 30°C using hot and cold water;

13. Repeat steps 1 to 12 using *Scenedesmus subspicatus* instead of *Chlorella vulgaris*.

14. After which, the data for both algae can be compared - the algae that gives the higher rate of photosynthesis at a lower light intensity has the lower light saturation point; (conclusion required to get full marks)

F. Annotated diagram

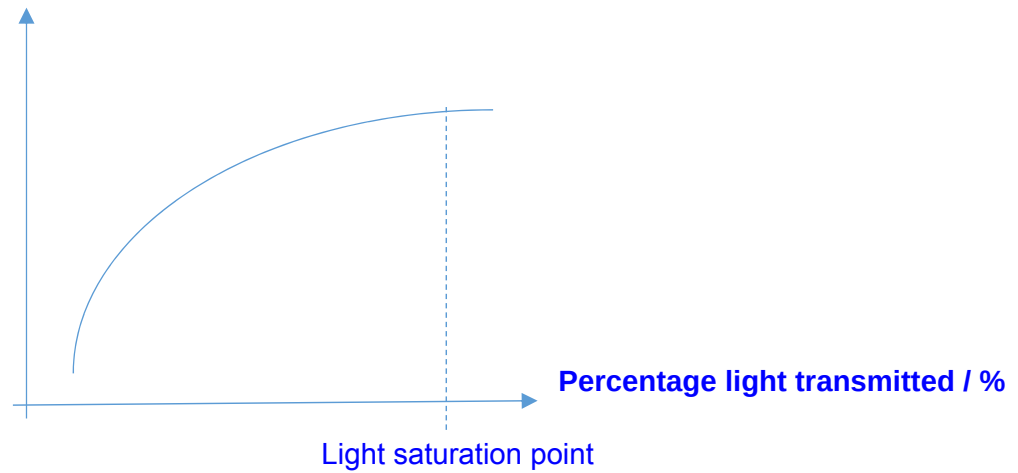
G. Data Recording and Processing (max 2)

1. Table

Table showing the rate of photosynthesis at different light intensities

Algae species	Percentage of light transmitted / %	Time taken for bromothymol blue to change from green to blue / s				Rate of photosynthesis / s ⁻¹
		Reading 1	Reading 2	Reading 3	Average	
<i>S. subspicatus</i>	20					
	40					
	60					
	80					
	100					
<i>C. vulgaris</i>	20					
	40					
	60					
	80					
	100					

2. Graph

Rate of photosynthesis / s^{-1} 

To modify graph: to include another curve for 2nd algae

R: straight line graph

A: start from 0 light intensity if they a setup in the dark was included in procedure

A: if never start from 0 even if there is a set up in the dark

H. Risks and Precautions (max 1)

1. Bromothymol blue may be a skin irritant. Wear gloves and goggles when handling it.

- (b) Microalgae such as *Chlorella vulgaris* has many uses. They can serve as bioremediation agents, as feed for aquaculture, as food for humans and animals, in pigment production, in bio-removal of heavy metals and in agriculture. It is usually desirable to establish large populations of microalgae, especially in aquatic environments where they are often employed.

One means of increasing microalgal populations may be to inoculate them with other microorganisms, such as *Azospirillum brasilense*, which is known as plant-growth-promoting bacteria.

To assess the effect of the *Azospirillum brasilense* on the proliferation of *Chlorella vulgaris*, *Chlorella vulgaris* was grown with and without *Azospirillum brasilense*. The number of algal cells was counted, with 10 samples taken from each of the two culture systems every two days over a period of 10 days in total.

Table 2.1 shows the results of the investigation.

Table 2.1

Condition <i>Chlorella vulgaris</i> was grown in	Mean number of cells in 1 cm ³ at the start	Mean percentage increase in number of cells of <i>Chlorella vulgaris</i>				
		day 2	day 4	day 6	day 8	day 10
<i>Chlorella vulgaris</i> grown without <i>Azospirillum brasilense</i>	10035	40	75	140	175	220
<i>Chlorella vulgaris</i> grown with <i>Azospirillum brasilense</i>	10108	65	125	195	280	330

A *t*-test was carried out to see if there is a significant difference between the number of cells in the two culture systems at 10 days.

The value of **t = 2.62** was obtained.

- (i) State a null hypothesis for this test.

.....[1]
 There is no significant difference in cell proliferation between *Chlorella vulgaris* grown in presence of *Azospirillum brasilense* and in absence of *Azospirillum brasilense*.

- (ii) State the number of degrees of freedom for this *t*-test.

.....[1]
 18

Table 2.2 shows some probability values of t .

Table 2.2

degrees of freedom	10	12	14	16	18	20	22	24	26	28	30	40	50	60
probability 0.05	2.23	2.18	2.14	2.12	2.10	2.09	2.07	2.06	2.06	2.05	2.04	2.02	2.01	2.00
probability 0.01	3.17	3.06	2.98	2.92	2.88	2.85	2.82	2.80	2.78	2.76	2.75	2.70	2.68	2.66

- (iii) Explain if the presence of *Azospirillum brasilense* in the growth medium is beneficial to the growth of *Chlorella vulgaris*.

.....

.....

.....

.....[2]

1. t value = 2.62, $df = 18$, $0.01 < p < 0.05$, hence $p < 0.05$ at 5% significance level,
2. the increase in cell number is significant and is not due to chance but is due to the presence of *Azospirillum brasilense*

[Total: 13]

3 **S1** is a slide of a stained transverse section through a plant stem.

You are not expected to be familiar with this specimen.

You are required to use a sharp pencil for drawings.

- (a) (i) Draw a large plan diagram of a sector of the stem **S1**, as shown by the shaded area in Fig. 3.1, to include **one complete vascular bundle**.

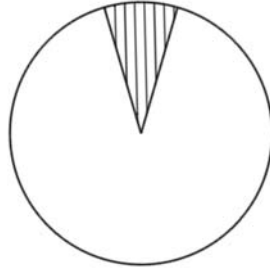
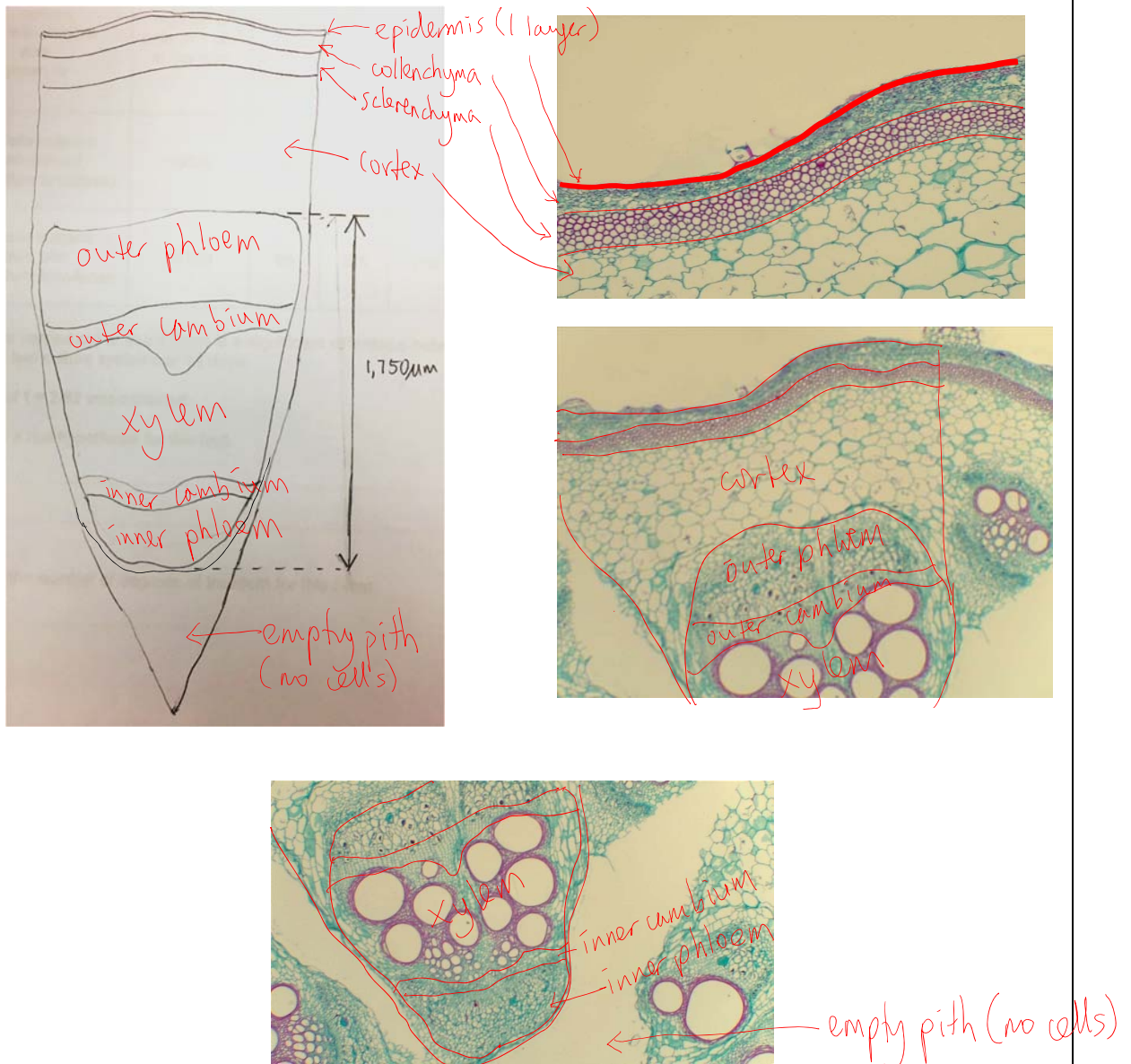


Fig. 3.1

State objective lens used: (ideal = 4X – 10X)



1. (D) Clear plan drawing with sharp, clear and unbroken lines;

Do not award marks if the following observed:

- Shading
- Ruled lines
- Overlapping lines
- 'Feathery' lines
- Less than 7 cm across length of drawing

2. (NC) No cells drawn

(TL) 4 tissues layers AND Proportion (epidermis thinnest, collenchyma and sclerenchyma about the same thickness, cortex the thickest)

3. (VB) one vascular bundle clearly shown within segment drawn;

Outermost line together with next 3 lines showing 'epidermis', 'collenchyma', 'sclerenchyma' and 'parenchyma' (students not expected to know cell types but should recognise the differences in the appearance of the cells)

~~(Do not award if there are less than 4 lines before vascular bundle outline)~~

4. (VBL) Vascular bundle drawn with at least 4 enclosed areas (total 5 areas - 'outer phloem', 'outer cambium', 'xylem', 'inner cambium' and 'inner phloem',)
5. (T) tapering shape of vascular bundle from outer phloem to inner phloem (NOT symmetrically oval);

- (ii) Use a ruler to measure your plan drawing and draw a line to show where you took the measurement. Now measure the actual size of the region where you took the measurement. Calculate the magnification of your drawing.

Show your working.

You may lose marks if you do not show clear working.

Example

Length of line on drawing = 8 cm = 80 mm = 80 000 μ m

Actual length of line = 1,750 μ m

Magnification = $80\,000 / 1750 = 45.7$ or 46 (rounded up to nearest whole number)
= X 46

Magnification =

[2]

1. Line included in plan drawing + actual dimension;
2. Division of drawing size by actual size and magnification given;

- (b) A student working on a slide similar to S1 wanted to calibrate her eyepiece graticule for use at a particular objective lens. Fig. 3.2 shows a diagram of a stage micrometer scale that she used to calibrate the eyepiece graticule.

One division, on either the stage micrometer scale or the eyepiece graticule, is the distance between two adjacent lines.

The length of one division on this stage micrometer is **0.1 mm**.

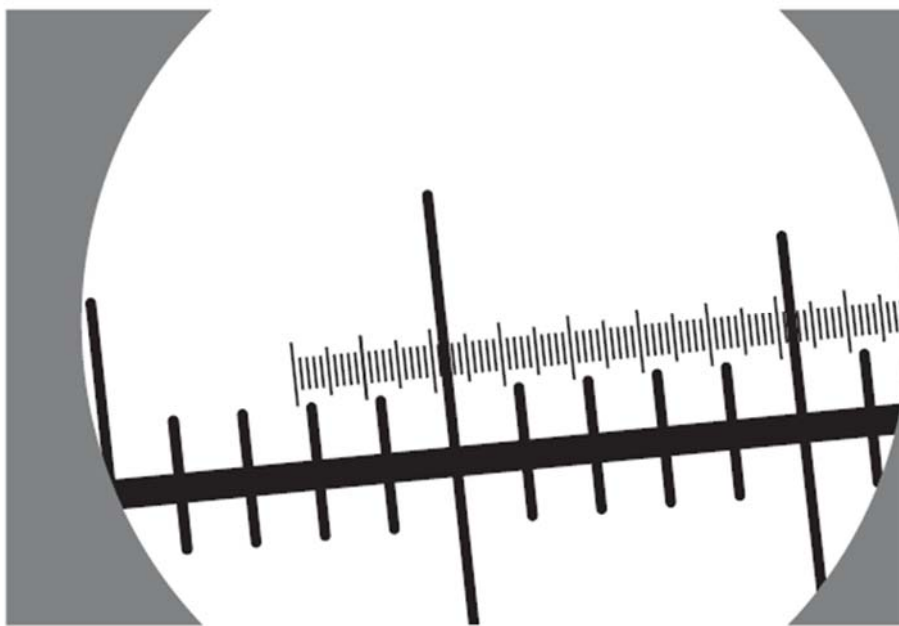


Fig. 3.2

- (i) Using Fig. 3.2, calculate the actual length of one eyepiece graticule unit by following the following steps.

Step 1

Length of 1 eyepiece graticule unit = mm divided by = mm

Step 2

Convert answer from Step 1 to a unit that is more suitable for use in light microscopy:

(answer from step 1) multiplied by =

[2]

1. 0.1 divided by 10 = 0.01 mm;
2. 0.01 multiplied by 1000 = 10 μm ; (R: no units in μm)

Fig. 3.3 is a photomicrograph showing a part of plant organ from another plant taken using the same objective lens from the same microscope in Fig. 3.2.

- (ii) Use the calibration of the eyepiece graticule from (b)(i) and **Fig. 3.3** to calculate the actual length of the plant tissue from X to Y.

You will lose marks if you do not show all steps in your calculation and if you do not use the appropriate units.

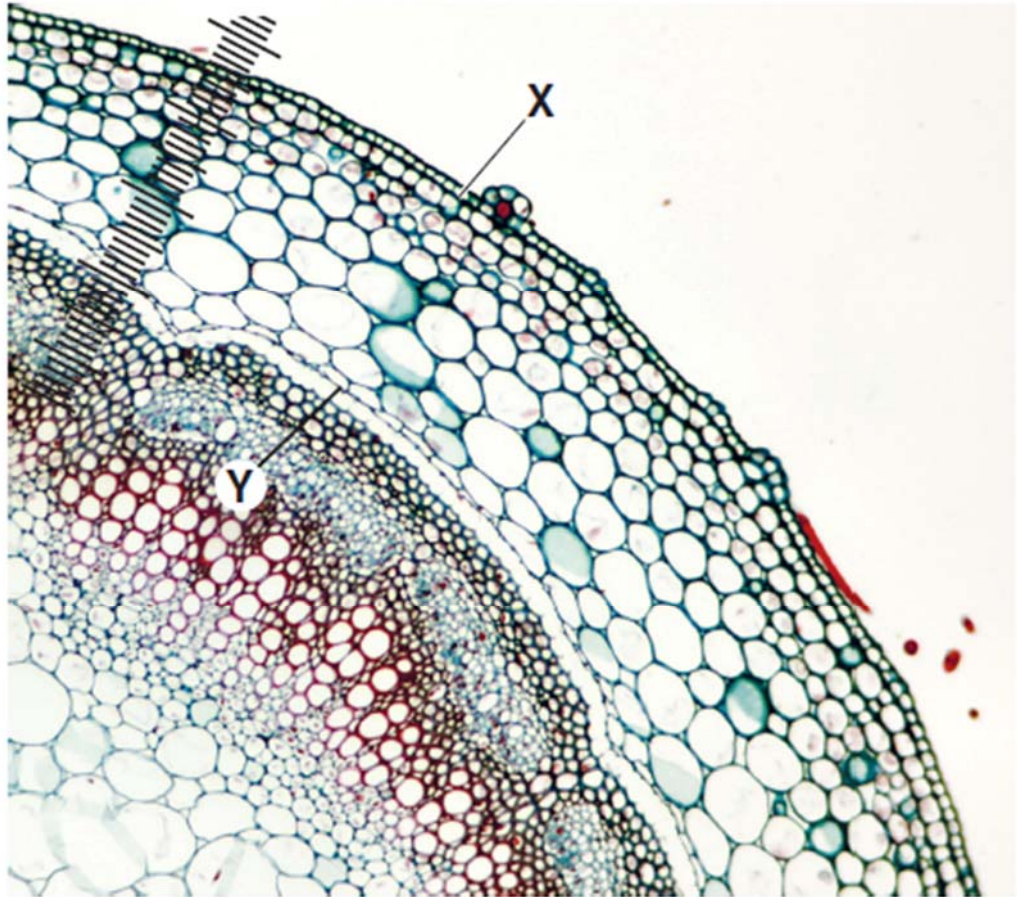


Fig. 3.3

Working:

1. Number of eyepiece graticule units along XY = 25; (accept: 23 - 25)
2. Length of XY tissue layer = 25 x 10 = 250 μm ; (accept: 230 - 250 μm)

[2]

- (c) Fig. 3.4 is a photomicrograph of a vascular bundle similar to that in slide **S1**. Using slide **S1**, locate and draw **five adjacent** circular structures in the region indicated by the label **T** in Fig. 3.4. Label each of these five structures **A** to **E**.

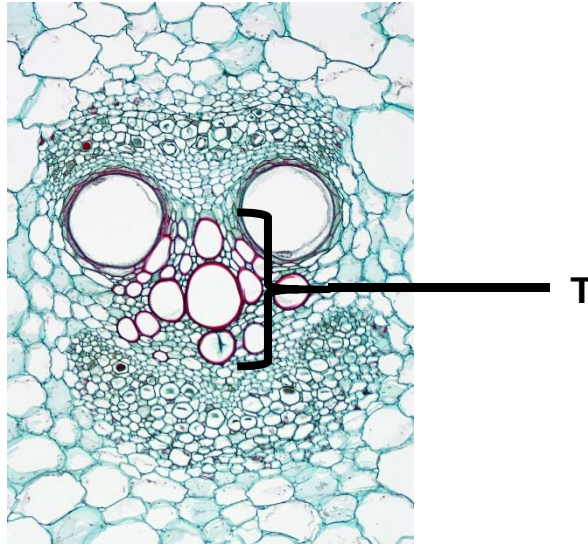
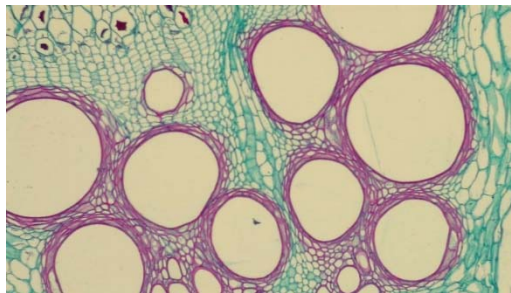
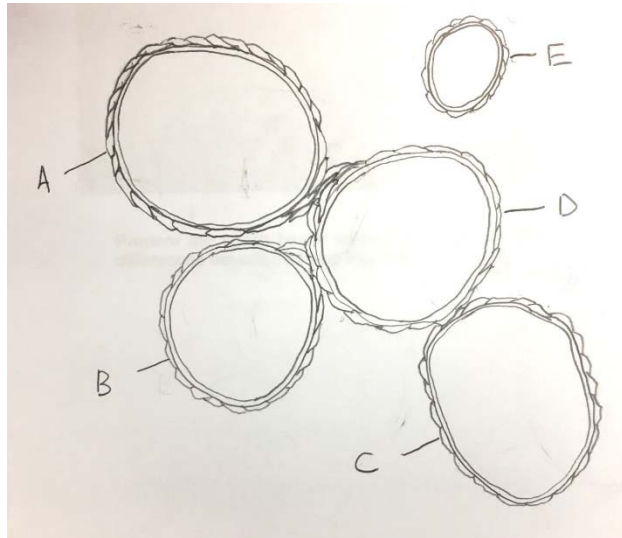


Fig. 3.4

State the objective lens used: (x4 – x10)



[5]

1. **(D)** Clear drawing with sharp, clear and unbroken lines;

Do not award marks if the following observed:

- Shading

- Ruled lines
 - Overlapping lines / 'breaks' in cells
 - 'Feathery' lines
 - 3 to 4 cm or less across widest cell (outermost line to opposite outermost line)
 - Drawn / trace over following outlines / distribution of cells from Fig. 3.4.
2. (FL) Only 5 cells drawn AND labelled A to E; (Do not award if less than 5 outlines drawn or no labels given)
 3. (R) Cells with rounded shape and without sharp corners; (Do not award if any cell shows sharp corners)
 4. (O) No organelles drawn (xylem lumen is empty); (Do not award if organelles drawn)
 5. (DL) Each cell has double lines around; (Do not award if inner line overlap and cross outer line)
 6. (WP) Wall pattern shown between two cells that are touching each other; (Do not award if total distance between three lines is more than 4 mm)

- (d) Use the eyepiece graticule to measure the diameter of each of the five circular structures you have drawn.

Calculate the mean diameter of the circular structures and the standard deviation (s).
Calculate standard deviation using the formula:

$$s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$$

Σ = 'sum of' $\bar{x}$ = mean
 n = sample size (number of observations)

Record your measurements and the results of your calculations in the space below.

You may lose marks if you do not show clear working.

Example

Mean of circular structures = $(250 + 180 + 300 + 200 + 100) / 5 = 206 \mu\text{m}$

$$s = \sqrt{\frac{(250-206)^2 + (180-206)^2 + (300-206)^2 + (200-206)^2 + (100-206)^2}{(5-1)}}$$

$$= \sqrt{\frac{44^2 + 26^2 + 94^2 + 6^2 + 106^2}{4}}$$

$$= 75.4 \mu\text{m} \text{ (3 s.f.)}$$

Mean diameter of circular structures =

Standard deviation of circular structures =

[2]

1. Working for mean diameter;

2. Working for standard deviation with proper units in μm + 3 s.f;

(e) Fig. 3.5 is a photomicrograph showing a stem from another plant.

You are not expected to be familiar with this specimen.

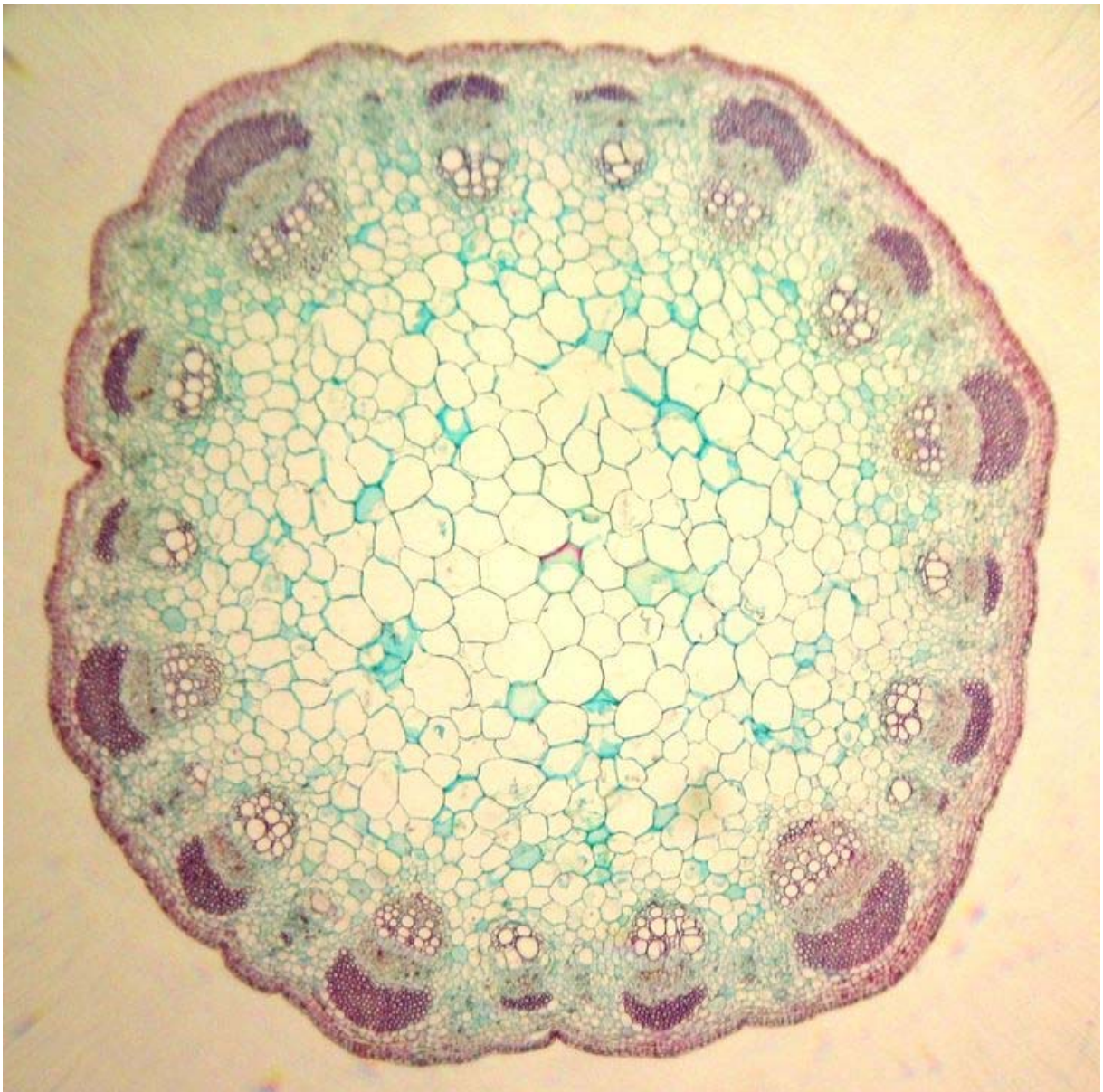


Fig. 3.5

Prepare the space below so that it is suitable for you to compare **three** observable differences between **S1** and **Fig. 3.5**.

[3]

S1	Fig. 3.5
1. <u>Angular</u> stem shape in t.s.;	<u>Circular</u> stem shape in t.s.
2. <u>Two rings</u> of vascular bundles	<u>Single ring</u> of vascular bundles
3. Vascular bundles alternate between <u>large and small</u> bundles between each ring;	Vascular bundles mostly of <u>similar sizes</u>
4. Vascular bundles have <u>more layers</u> of tissues compared to those of Fig. 3.5;	Vascular bundles have <u>less layers</u> of tissues compared to vascular bundles of S1
5. <u>Empty</u> central region (empty pith);	<u>Filled</u> central region (pith present)

End of Paper